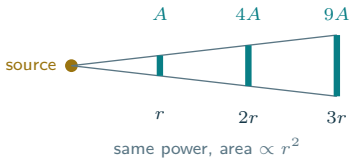
RECIPROCALS ON ONE PAGE

The reciprocal-power hinge

From potential as $1/r$ to field strength and flux density as $1/r^2$, with geometry setting the power.

1 / THE RULE AND THE PICTURE

A point source spreads over a sphere whose area grows as r^2



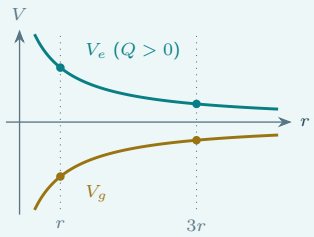
A point source gives two related radial patterns, with the constants and signs set by the source and the power of r set by radial geometry (point source or spherical symmetry, no absorption):

$$\text{potential} \propto \frac{1}{r}, \qquad \text{field strength, flux density} \propto \frac{1}{r^2}$$

Potential: energy per test property. **Field**: force per test property. **Irradiance**: power per unit area.

2 / POTENTIAL / ONE RECIPROCAL POWER

$V_g = -GM/r$ and $V_e = kQ/r$ share the structure $1/r$

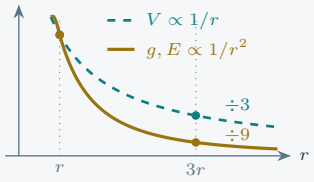


For a point mass and a point charge, $V_g = -GM/r$ and $V_e = kQ/r$ are energies per unit test property, E_p/m and E_p/q, with zero at infinity. Moving from r to $3r$ divides $|V|$ by 3: because V_g is negative this is a rise towards zero, whereas the positive charge's V_e falls towards zero. The numerator carries source strength and coupling, and the sign records gravity's attraction or the source charge. For Earth, V_g rises from $-62.6\ \mathrm{MJ\,kg^{-1}}$ at the surface to $-20.9\ \mathrm{MJ\,kg^{-1}}$ at $3R_E$.

$$|V(3r)| = \tfrac{1}{3}|V(r)|, \qquad V_g \text{ rises towards } 0 \text{ from below}$$

3 / FIELD STRENGTH / TWO RECIPROCAL POWERS

$g = GM/r^2$ and $E = k|Q|/r^2$ fall by nine when the distance triples

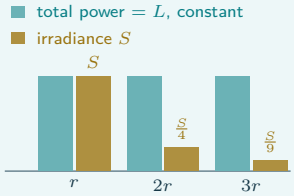


For the same sources $g = GM/r^2$ and $E = k|Q|/r^2$, with $F_g = mg$ and $F_e = qE$, so tripling the distance divides both by 9 while the potential falls only by 3: for Earth, $9.8\ \mathrm{m\,s^{-2}}$ at the surface becomes $1.1\ \mathrm{m\,s^{-2}}$ at $3R_E$. Potential is scalar energy per unit test property and field strength vector force per unit test property, so "both decrease with distance" misses the distinction. **E** points away from positive and towards negative charge, whereas g points towards the source.

$$g(3r) = \tfrac{1}{9}g(r), \quad E(3r) = \tfrac{1}{9}E(r), \quad V(3r) = \tfrac{1}{3}V(r)$$

4 / SPHERICAL SPREADING / FLUX DENSITY

Irradiance falls as $1/r^2$ while the power through each sphere stays at L



A source of power L radiating equally in all directions spreads it over a sphere of area $4\pi r^2$, so the irradiance, or power per unit area, is $S = L/(4\pi r^2)$. Doubling the distance quarters S, and for $L = 3.8 \times 10^{26}$ W at $r = 2.0 \times 10^{11}$ m, $S \approx 7.6 \times 10^2\ \mathrm{W\,m^{-2}}$. Because $S(4\pi r^2) = L$ on every sphere, the total power through each complete sphere is unchanged while the irradiance falls, since the same power meets an area growing as r^2.

$$S = \frac{L}{4\pi r^2} \approx 7.6 \times 10^2\ \mathrm{W\,m^{-2}}, \qquad S\,(4\pi r^2) = L$$

5 / SYNTHESIS

The $1/r^2$ expresses spherical spreading; the numerator supplies the source's physics

quantity	dependence	$r \to 2r$	what it measures
potential	$1/r$	$\times\frac{1}{2}$ in magnitude	energy per unit test property
field strength	$1/r^2$	$\times\frac{1}{4}$	force per unit test property
irradiance	$1/r^2$	$\times\frac{1}{4}$	power per unit area
total power through a sphere	constant	unchanged	the source power L

ASIDE / WHY POTENTIAL HAS ONE POWER FEWER

Energy is force times distance, so $V = \int_r^\infty g\,\mathrm{d}r' = GM/r$ loses one power of r on integrating, and conversely $g = -\mathrm{d}V/\mathrm{d}r$.

ASIDE / FOUR MISCONCEPTIONS

Potential and field are not interchangeable; negative potential rises towards zero; flux density and total flux differ; inverse-square needs a point source and no absorption.

Convention. After worksheet *The Reciprocal-Power Hinge*; check in Python: `3.8e26/(4*pi*(2e11)**2)` gives 755.99. Companion to momentum and torque hinges.

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